

OSSP:

Commands:

Uname : Displays system information.

lscpu : Shows CPU details.

```
[shashankreddy@Shashanks-MacBook-Air ~ % ./task
Enter a Linux command: uname

Parent Process
Parent PID : 21284
Child PID  : 21285

Child Process
Child PID : 21285
Parent PID: 21284

Executing command: uname

Darwin
```

ps : Displays the currently running processes.

```
shashankreddy@Shashanks-MacBook-Air ~ % ./task
Enter a Linux command: ps

Parent Process
Parent PID : 21288
Child PID  : 21289

Child Process
Child PID : 21289
Parent PID: 21288

Executing command: ps

  PID TTY          TIME CMD
 21273 ttys000    0:00.02 -zsh
 21288 ttys000    0:00.01 ./task
```

Relation Between Hardware Resources and Operating System Services:

1. CPU

The operating system shares the CPU among different programs so they can run together.

Example: Chrome and a music player both receive CPU time.

2. Memory (RAM) → Memory Management

The operating system provides separate memory space for each running program.

Example: Chrome and VS Code use their own memory.

3. Storage (SSD) → File System Management

The operating system manages and organizes files on the storage device.

Example: Saving a file in the **Documents** folder.

4. I/O Devices → Device Management

The operating system controls input and output devices using device drivers.

Example: Using a printer or keyboard through the operating system.